PashuDrishti AI ([redacted])

HACKATHON DEFENSE & JUDGES Q&A; PREPARATION GUIDE

Problem Statement PS-5: AI-Driven Cattle & Buffalo Breed Identification | Bharat Pashudhan Mission

PART 1: Tech Stack Defense — Why Our Architecture Beats Alternatives

Component	What We Used	Why It Beats Alternatives (TensorFlow, Flask, ViT, React)
Model Backbone	PyTorch ResNet-18 (Weights: ~44MB)	**Over Vision Transformers (ViT) & ResNet-101:** ViT takes 2.5s+ and 350MB+ memory on CPU. ResNet-18 has 11.2M params, executes in <250ms on CPU, and avoids severe overfitting on small-to-medium Indian cattle datasets.
Inference Pipeline	Two-Tier Hierarchical Classification	**Over Single-Step Classifiers:** Single classifiers try to assign a breed to anything (cars, humans, dogs). Tier-1 validates Cow vs Buffalo vs Non-Bovine (55% threshold) first, preventing hallucinations.
Backend API	FastAPI + Uvicorn (ASGI Python 3.12)	**Over Synchronous Flask/Django:** 200-300% faster throughput. Asynchronous event loop handles concurrent photo uploads from field workers without blocking CPU threads.
Frontend	Vanilla ES6 + High-Contrast Agritech CSS	**Over React / Next.js / Vue:** Zero JS bundle bloat (~35KB total). Loads in <200ms on 2G/3G rural networks. Direct access to HTML5 MediaDevices camera without third-party package conflicts.
Deployment	Render Cloud (Docker Container Edge)	**Over Streamlit/Local Ngrok:** 24/7 permanent URL with global edge SSL. Does not drain laptop battery and scales reliably for live judge evaluations.

PART 2: Top Technical Questions Asked by Judges & Winning Answers

Q1: Why did you choose ResNet-18 over heavier architectures like ResNet-50, YOLOv8, or Vision Transformers (ViT)?

Winning Answer: 'That was a deliberate engineering decision based on rural deployment constraints. First, **Inference Latency:** ResNet-18 has only 11.2 million parameters and executes on a standard 2-vCPU cloud container in under 250 milliseconds. Vision Transformers require GPUs or take 2 to 3 seconds on CPU, draining smartphone battery. Second, **Overfitting Resistance:** Fine-grained Indian livestock datasets across 41 classes are prone to overfitting on large models. ResNet-18 offers the optimal balance of high feature extraction capacity, compact weights (44MB), and sub-second execution.'

Q2: What happens if an enumerator accidentally uploads a photo of a tractor, dog, or human? Does your AI guess a cow breed?

Winning Answer: 'No, and that is our primary architectural safeguard: our **Dual-Tier Triage Pipeline**. Tier 1 passes the image to a 3-class discriminator (Cow, Buffalo, None). If the image is non-bovine or if the species confidence falls below our calibrated 55% threshold, the pipeline halts immediately with a clear alert. The 41-breed model is never invoked on invalid imagery, eliminating false hallucinations.'

Q3: How do you mathematically compute the Top-3 breeds?

Winning Answer: 'We take the 41-dimensional raw output logits from ResNet-18 and apply the **Softmax function**: *P(Breed_i) = exp(z_i) / sum(exp(z_j))* to produce a true probability distribution. We then invoke *torch.topk(probabilities, k=3)* to extract the top three highest probabilities and their class indices. This satisfies Problem Statement PS-5 by replacing one rigid, brittle answer with a transparent, ranked confidence breakdown.'

Q4: How do you protect the system against malicious uploads or server crashes?

Winning Answer: 'We implemented multi-layered API defenses in FastAPI: (1) **10MB Payload Cap** to prevent DoS attacks; (2) **MIME-type whitelist** (JPEG/PNG/WEBP only); (3) **Decompression Bomb Protection** clamping Image.MAX_IMAGE_PIXELS to 25 million pixels; (4) **Parameter Sanitization** restricting query values between 1 and 5.'

PART 3: Gov-Tech & Bharat Pashudhan Domain Questions

Q5: Why is Top-3 prediction better for Bharat Pashudhan than a single definitive answer?

Winning Answer: 'Livestock genetics in Indian villages are rarely black-and-white due to historical crossbreeding. For example, a Sahiwal cow from the Montgomery tract shares over 75% phenotypic traits with a Red Sindhi cow. If an AI gives one rigid 51% guess, an untrained worker has no choice but to accept it. With a Top-3 ranking, the worker sees: #1 Sahiwal (72%) | #2 Red Sindhi (21%) | #3 Gir (4%). Paired with our on-screen visual hallmark checklist (ear curve, dewlap, horn orientation), the worker verifies the physical animal with their own eyes before entering verified data.'

Q6: Why did you implement 5 regional languages instead of just English and Hindi?

Winning Answer: 'India’s bovine diversity is intensely regional: Gir and Kankrej in Gujarat, Murrah in Punjab/Haryana, Khillari and Dangi in Maharashtra, and Sahiwal across the northern plains. Grassroots enumerators and dairy farmers are native speakers who think in their regional mother tongues. Forcing an English-only interface causes hesitation and data-entry errors. Our app translates all UI elements, breed names (e.g., ■■■■■■■ (Sahiwal)), and anatomical terms natively into English, Hindi, Bengali, Gujarati, and Marathi.'

Q7: How does this project prevent financial waste in government livestock schemes?

Winning Answer: 'Subsidies for artificial insemination (AI) and indigenous breed conservation under the Rashtriya Gokul Mission are breed-specific. Misclassifying non-descript cattle as high-pedigree Gir or Sahiwal wastes public funds. PashuDrishti AI provides an auditable, downloadable verification slip with timestamped certainty scores, ensuring financial assistance reaches genuine farmers.'

PART 4: Countering Tough 'Gotcha' Questions (Trap Questions)

Q8: 'Isn't your model just classifying based on background grass and lighting rather than the animal?'

Winning Rebuttal: 'We specifically addressed spatial bias during training. First, our standard ImageNet normalization (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) balances chromatic variations. Second, our dataset incorporates cattle across dirt pens, muddy paddy fields (Dangi), arid sand tracts (Tharparkar, Rathi), and clean dairy farms (Holstein Friesian). The deep convolutional filters in ResNet-18 learn anatomical edge invariants—such as the lyre-shaped horns of Kankrej and the pendulous dewlap of Sahiwal—not background grass.'

Q9: 'What if there is no internet connectivity in remote tribal villages?'

Winning Rebuttal: 'Our architecture was designed with an offline-first roadmap in mind. The current web version is optimized for low-bandwidth 2G/3G networks with a total bundle size of ~35KB and client-side canvas resizing. Furthermore, because our trained PyTorch model is compact (44MB), our next milestone converts the weights into an **ONNX Runtime / TensorFlow Lite** package embedded directly into the Android application, enabling 100% offline edge inference.'

Q10: 'Why didn't you build this with Streamlit or Gradio?'

Winning Rebuttal: 'Streamlit is an excellent rapid prototyping tool, but it is unsuitable for government field deployments. Streamlit re-executes the entire script on every user interaction, leading to high server CPU usage and crashes under concurrent users. It also struggles with responsive hardware camera controls on mobile browsers. We built an enterprise ASGI stack with FastAPI and lightweight Vanilla JS, delivering sub-200ms page loads, camera shutter support, and full multilingual reactivity.'